Networking Lab Cheat Sheet

Add user:

sudo adduser rashik	Adds user
sudo usermod -aG sudo rashik	Gives user more power

IP Section:

Command	Explanation
ip a	Displays all network interfaces with their **IP addresses, MAC addresses, and status**.
ip a show dev eno2	Shows detailed IP configuration **only for the network interface eno2**.
sudo ip a add 172.17.100.19/24 dev eno2	Assigns the IP address **172.17.100.19** with subnet mask **/24** to interface **eno2**.
sudo ip a del 172.17.100.19/24 dev eno2	Removes the assigned IP address from interface **eno2**.
ping 10.34.0.142	Sends continuous ICMP packets to test connectivity to the destination.
ping –c 5 10.34.0.142	Sends **only 5 ICMP packets** then stops.
ping –n –c 5 10.34.0.142	Sends 5 packets and **does not resolve hostnames** (shows numeric IP only).
ping –n –c 5 –R 10.34.0.142	Sends 5 packets, does not resolve names, and **records router path (route tracing)** if supported.
ping –I 172.17.100.19 10.34.0.142	ping –I <source> <destination>

ip r	Displays the **current routing table**
sudo ip r add 172.17.100.19/32 via 10.34.0.143 dev eno2	Adds a **route to a specific host (172.17.100.19)** through gateway **10.34.0.143** using network interface **eno2**.
sudo ip r del 172.17.100.19/32	Deletes the **specific routing rule** for that host from the routing table.
ip neigh	Shows the **ARP table** (maps IP addresses to MAC addresses of nearby devices).
ipcalc 172.17.100.19/24	Shows ip calculation of this host and some other stuff

SSH Server:

Command Sequence	Purpose
whoami	Shows username
sudo apt install openssh-server	Does what it says
which sshd	Shows the location of the sshd executable file in the system path.
sudo systemctl start sshd	Starts the ssh server for remote connection
sudo systemctl status sshd	Shows server status
sudo systemctl disable --now ssh.socket	Disable ssh.socket for ssh to work normally
cd /etc/ssh	Go to the ssh config directory
sudo nano sshd_config	Uncomment the port, and change it e.g Port 2233 (Dont change if not asked)
sudo systemctl restart sshd	Restarts ssh service
ssh –p 2233 username@ip	Connect to SSH using a **custom port**
sudo reboot	
logout	Closes connection

sudo scp -P 2233 iit-du@10.100.202.234:~/Desktop/file1.txt ~/Desktop/file2.txt	To transfer a file from remote pc to user pc Enter the port and user name of the remote pc and ensure that file1.txt exists in remote pc
sudo scp -P 2233 ~/Desktop/file1.txt iit-du@10.100.202.234:~/Desktop/file2.txt	To transfer a file from user pc to remote pc Enter the port and user name of the remote pc and ensure that file1.txt exists in user pc

Misc	
sudo ss -tulnp | grep ssh	If ssh is saying connection is refused, then run this and check the port it is currently listening to and use that port
sudo nano /etc/ssh/sshd_config	If the correct password is being refused or denied, then adjust in the following way

Comment the following lines like this

kbdinteractiveauthentication

passwordauthentication

<u>Bind:</u>

Command Sequence	Purpose
sudo apt install bind9	Duhhh
which named	Shows the **location of the DNS server daemon (named) executable** in the system.

sudo systemctl start bind9	Start DNS Service (Ubuntu)
sudo systemctl status bind9	Check Status
cd /etc/bind/	Go to bind directory
sudo cp db.local db.s19.zhjlab.bd	Copy
sudo nano db.s19.zhjlab.bd	Change stuff here first

Change it to

```
$TTL 300
@       IN      SOA     ns1.s19.zhjlab.bd. bsse1619.iit.du.ac.bd. (
                        32 ; Serial (incremented)
                        604800 ; Refresh
                         86400 ; Retry
                         2419200 ; Expire
                        604800 ) ; Negative Cache TTL

; Nameserver
@                   IN  NS      ns1.s19.zhjlab.bd.
ns1.s19.zhjlab.bd.  IN  A       172.17.100.19
@                   IN  A       172.17.100.19
www                 IN  CNAME   ns1.s19.zhjlab.bd
```

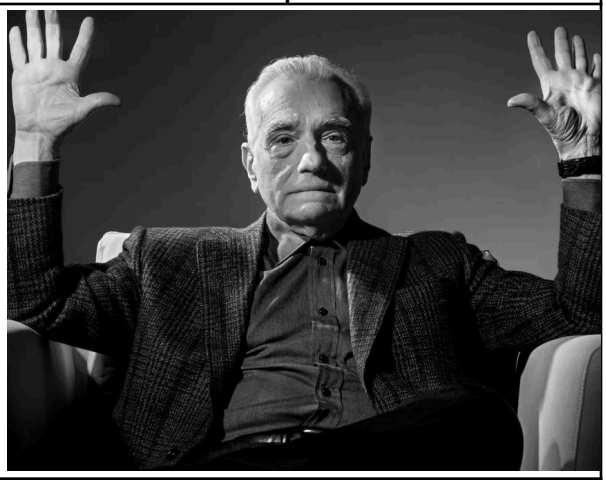
sudo nano /etc/bind/named.conf.local	Add here

```
zone "s19.zhjlab.bd" {
    type master;
```

```
    file "/etc/bind/db.s19.zhjlab.bd";
};
```

sudo named-checkzone s19.zhjlab.bd /etc/bind/db.s19.zhjlab.bd	Check zone file If “Ok” proceed
sudo tail -f /var/log/syslog	Run this in another terminal tab and monitor this (optional)
sudo systemctl restart bind9	Restart bind service
dig @172.17.100.19 s19.zhjlab.bd	Check (Should get Answer 1)

Apache:

Command Sequence	Purpose
sudo apt install apache2	
sudo systemctl start apache2	Starts the apache service
localhost	Type in browser (verification)
cd /var/www/html	Go to document root
sudo nano index.html	Write an html file

```
<html>
<head>
<title>OOOOOOOOOOOOOOOOOOOOOOO</title>
</head>
```

```
<body>
<h1>UUUUUUUUUUUUUUUUU</h1>
<p>I wanna die</p>
</body>
</html>
```

sudo systemctl restart apache2	Restarts the service
sudo mkdir /var/www/mysite	Creates mysite directory
sudo cp /var/www/html/index.html /var/www/mysite/index.html	Copies the index html file to mysite directory
sudo cp /etc/apache2/sites-available/000-default.conf /etc/apache2/sites-available/001-mysite1.conf	Copy the config file
sudo nano /etc/apache2/sites-available/001-mysite1.conf	Edit

Change ServerName part to

ServerName s19.zhjlab.bd

and

Change document root from

/var/www/html

to

/var/www/mysite

sudo nano /var/www/mysite/index.html	Change some stuff here
sudo a2ensite 001-mysite1.conf	Enable the site
sudo a2dissite 000-default.conf	Better to disable the default site(optional)

sudo nano /etc/hosts	Edit

Add
172.17.100.19 s19.zhjlab.bd

sudo systemctl restart apache2	Restarts the service
http://s19.zhjlab.bd	For verification

If you want to see a remote website(http://s00.zhjlab.bd) configured by someone else with ip 172.17.100.119 from your machine

sudo ip r add 172.17.100.119/32 via 10.34.0.143 dev eno2	First we have to add a route for the remote ip 172.17.100.119 so that we can ping it Here 10.34.0.143 is your gateway ip
ping s00.zhjlab.bd	should resolve to 172.17.100.119 (verification)
sudo nano /etc/hosts	Edit

Add

172.17.100.119 s00.zhjlab.bd

sudo systemctl restart apache2	
Close the browser	
Open the browser	
http://s00.zhjlab.bd	Should work now